

Functional Design Specification

Price Book Automation

April 2026

1. Executive Summary

AudioCodes publishes a comprehensive suite of Price Books every quarter, spanning global channel partners, North America distributors, and Annual Contract Maintenance (ACM) customers across multiple currencies and regions. This document presents the validated scope, measured impact, and phased delivery plan for automating this process end to end.

2. The Business Challenge

Each quarterly cycle is a high-stake, multi-week coordination effort. The data from Q2-26v1 reveals the scale of the problem: from a base of 3,854 CPN lines, only 100 lines changed this quarter - yet the entire pipeline of 9+ output files, 3 currencies, and 2 regions had to be regenerated, reviewed, and distributed manually. The ratio of effort to change is disproportionately high.

Disproportionate Effort	Only 2.6% of CPN lines changed in Q2-26v1, yet 100% of files required manual regeneration, validation, and distribution. The pipeline cannot selectively process only what changes.
Manual Dependency	Every step - QA validation, macro execution, currency conversion, ACM generation, NA uplift - depends on individual contributors following documented procedures. A missed step cascades through all downstream files.
Error-Prone Handoffs	Content is copied between spreadsheets by hand. The process docs explicitly warn PMs: 'Do not copy lines from other files,' 'Do not add data validations.' These rules exist because violations are common.
ACM Script Fragility	The Python ACM script uses deprecated pandas methods (df.append) and hardcoded product-to-abbreviation mappings for 20+ product lines. Adding a new product requires code changes, not configuration.
Scale Constraints	Adding new regions, currencies, or product lines requires more manual work. The distributor discount VLOOKUP, NA uplift logic, and ACM currency variants all multiply linearly with scope.
No Automated Audit	Version Updates sheets exist in 3 separate files (General, NA, ACM) with no cross-validation. Tracing a pricing discrepancy requires manual inspection across multiple workbooks.

3. Current Process: End-to-End Flow

The pipeline below is validated against the actual process documentation. Every step shown in green is currently active and functioning - but executed manually by the team each quarter.

Legend

- GREEN (Phase A) - priority: live QA on PB Draft, real-time validation as the manager edits
- Current process steps are active today but executed manually each quarter
- YELLOW (Phase B) - Automation of existing steps: same process, zero manual effort
- RED (Phase C) - Transformation: move beyond Excel into a robust, integrated platform

3.1 Process Stages

Stage 1 PM Input	PMs edit the shared PM PB Excel file (co-authoring via desktop app with auto-save). Each PM updates their product section: adding new CPNs, updating prices/descriptions, or marking lines for deletion. The Editing Status tab tracks who have completed their section.
Stage 2 QA Gate	15 manual validation rules are executed before the file can be handed off. These include CPN format checks (max 20 chars, no spaces, uppercase), duplicate detection, mandatory field validation, EUR price accuracy (MSRP USD ÷ D10 rate), Notes keyword scan (no % or Discount references), external link detection, and merged cell checks.
Stage 3 Macro PB Build	PM PB content is copied into the Macro PB Master file. Cover tab updated with version, dates, and USD/EUR rate. Version Updates tab constructed from New/Update/Delete lines. EOS tab updated. Master tab cleaned: Delete lines removed, Status/Remarks columns deleted, font color rules applied.
Stage 4 Macro Execution	VBA macro in the .xlsm file generates all ROW Channel Price Books: ACTS USD, ACTS EUR, CHAMPS USD, CHAMPS EUR. Post-generation, Eyal applies manual cross-category edits: removing CHAMPS lines from ACTS files and ACTS/AHR lines from CHAMPS files per documented scope rules.
Stage 5 ACM Generation	Python script (ACM_PB_creation_main.py) maps CAPEX CPNs to ACM items using a ±2% MSRP deviation matching rule. Generates ACM Version Updates. Build ACM_USD_PRICEBOOK and Uploadable Flat tabs. Sent to Meital for approval. Currency variants (EUR, GBP, AUD) are created from an approved USD base.
Stage 6 NA Price Books	ROW USD PB copied and renamed. NA uplift rules applied via VLOOKUP: +25% for HW/Accessories, +15% for M9K, +10% for TAA products ≤\$7,500. Device pricing reviewed by Nir for each quarter. NA Version Updates accumulated from v0 + v1 deltas.
Stage 7 Distributor PB	NA ACTS USD PB copied. Master Distributor Discount column added via VLOOKUP from reference file. Net prices are calculated. Special values handled: No Charge → \$0, Quotation → \$1, ACM-PB → as-is. Values pasted.

Stage 8 Review & Release

All files reviewed by stakeholders. Meital provides final ACM approval. Oracle updated: ACM relationship end dates, new ACM items with UOM, new ACM relationships. All files are distributed on the effective date (1st of quarter month).

3.2 File Hierarchy

#	File	Derived From	Purpose
1	PM PB (Q2-26v1)	PM PB Q2-26v0	Final PM input with all CPN changes
2	Macro PB (.xlsm)	PM PB v1	Master file; source of truth for all outputs
3	Channel PB - ACTS USD	Macro PB	ROW channel book with ACTS support pricing
4	Channel PB - ACTS EUR	Macro PB	ROW channel book, EUR currency
5	Channel PB - CHAMPS USD	Macro PB	ROW channel book with CHAMPS pricing
6	Channel PB - CHAMPS EUR	Macro PB	ROW channel book, EUR currency
7	NA Channel PB - ACTS USD	ROW ACTS USD + NA rules	North America with regional uplift
8	NA Master Distributor PB	NA ACTS USD + discounts	Distributor pricing with net price
9	ACM PB - USD	Macro PB + Python	Annual maintenance, ROW USD
10	ACM PB - EUR	ACM USD + rate	Annual maintenance, ROW EUR
11	NA ACM PB - USD	ROW ACM + NA uplift	NA annual maintenance pricing
12	Version Updates	Macro PB delta	CPN change log for the quarter
13	ACM Version Updates	Python script	ACM-specific change log

4. Phase A - Live QA on PB Draft

Before any automation of downstream steps, the priority is to protect the source. Today, when the manager edits the PB Draft file - changing the price, updating a CPN, modifying a description - there is no real-time feedback. Errors are only caught later, during the 15-step manual QA gate, after PMs have already built their changes on top of potentially incorrect data.

Phase A delivers a live validation layer directly to the PB Draft file. As the manager types, the system validates in real time: field format, pricing consistency, CPN uniqueness, description length, and pattern-based anomaly detection (e.g., if a PN ending in '_PS' typically costs \$10 more than its base, the tool flags any deviation immediately). This ensures PMs receive a clean, validated baseline before they begin their quarterly edits.

Principle: *Catch it at the source. Validate as you type, not after the fact.*

Phase A: Live QA Scope

- Real-time field validation on every edit: CPN format (max 20 chars, no spaces, uppercase), mandatory fields, description length
- Instant pricing alerts: flags unusual price changes compared to the previous version or peer products in the same category
- Pattern-based anomaly detection learns suffix/prefix pricing relationships (e.g., _PS = +\$10, /R = 1.5x base) and alerts on deviations
- Duplicate CPN detection as you type - before the line is even saved
- Live EUR price preview shows the calculated EUR price based on the current USD/EUR rate as the manager enters MSRP USD
- This is the first deliverable of the project - it protects every downstream step by ensuring the source data is clean from the moment it is entered

5. Phase B - Automating the Existing Process

Phase B retains the current process structure but replaces human coordination with automated execution. Every output remains identical; the effort to produce it changes fundamentally.

Principle: *Same process. Same outputs. Zero manual effort.*

Phase B Automation Scope

- Script all 15 QA validation rules into a single automated pre-release gate (CPN format, duplicates, field completeness, EUR accuracy, Notes keyword scan, external link detection, merged cells)
- Auto-detect ACM duplicate CPNs and flag HW/SW conflicts - currently the #1 manual intervention in every ACM cycle
- Apply NA regional uplift programmatically: +25% HW, +15% M9K, +10% TAA ≤\$7,500, with device pricing feed from Nir's quarterly file
- Automate Eyal's post-macro edits: cross-category line removal in ACTS/CHAMPS files per documented scope rules
- Sync Agile picklist values automatically (Discount Level and CPN Type) - eliminating the manual comparison step
- Normalize Version Updates status values (case-insensitive matching) and add cross-file consistency checks
- Automate post-macro font-color painting and column cleanup in the Master tab

6. Phase C - Transforming Beyond Excel

Phase C moves AudioCodes from an Excel-centric workflow - where business logic lives in VBA macros, spreadsheet formulas, and manual file operations - to a modern, platform-agnostic pipeline. The PB Draft General document already outlines the target entity model (PB Draft, Notes, Support Items, Range tables); Phase C builds the infrastructure to realize it.

Principle: *Transform the foundation. One source of truth. On-demand generation.*

Phase C Transformation Scope

- Replace the dual PM PB + Macro PB file model with a unified data model - a single, structured source for all CPN data, pricing rules, and version history
- Implement the PB Draft entity model: PB Draft records with auto-generated Support Items, Notes linked by Product, Range-based pricing lookups
- Build API-based integration with Agile (picklist sync, CPN Type validation) and Oracle (ACM relationships, end dates, UOM creation)
- Replace VBA macro with a code-based, parameterized ETL pipeline: region, currency, and output format as configuration, not manual steps
- Centralize all three Version Updates (General, NA, ACM) into a single audit log with automated cross-validation
- Decouple output format from process logic: generate Excel, PDF, or system uploads from the same pipeline

8. Intelligent Pricing Validation

Beyond automating the existing process, this initiative introduces a new capability that does not exist today: intelligent, data-driven pricing verification. This layer serves two purposes - visual tools for human verification, and AI-based anomaly detection that catches errors no checklist can anticipate.

8.1 Visual Pricing Verification Dashboard

Product Managers and reviewers currently have no efficient way to verify that pricing strategy was implemented correctly across thousands of CPN lines. The verification dashboard provides interactive tools to filter, compare, and visually inspect pricing decisions before release.

Dashboard Capabilities

- Filter and highlight specific product models, categories, or CPN types instantly - replacing manual scrolling through 3,854+ rows
- Side-by-side price comparison across versions (Q1 vs Q2, v0 vs v1) with visual diff highlighting for every changed CPN
- Price distribution charts by product group: scatter plots, histograms, and box plots that reveal outliers briefly
- NA uplift verification view: overlay ROW and NA prices to confirm the correct uplift percentage was applied per category (+25% HW, +15% M9K, +10% TAA)
- Currency consistency check: visual comparison of USD, EUR, and AUD prices to confirm conversion rates were applied uniformly
- ACM pricing scatter plot: CAPEX MSRP vs. ACM M36 pricing to verify the calculated maintenance ratios fall within expected ranges
- Export any filtered view or chart for stakeholder review or audit documentation

This capability transforms pricing QA from a reactive checklist ('did we follow the rules?') into a proactive intelligence layer ('does this look right given everything we know?'). The visual dashboard is delivered in Phase B as part of the QA automation tooling. The AI anomaly detection engine is delivered in Phase C as part of the integrated pipeline.

9. Phase Comparison

Dimension	Today (Green)	Phase B (Yellow)	Phase C (Red)
Source QA (PB Draft)	No validation at input	Live real-time field QA	Integrated into data model
QA Validation	15 manual checks	Automated script	Integrated CI gate
Pricing Verification	Manual scrolling	Visual dashboard + filters	AI anomaly detection
ACM Generation	Python + manual review	Refactored, automated	API-integrated
Agile Alignment	Manual comparison	Auto-sync	Live API binding
Version Updates	3 separate manual files	Auto cross-validated	Unified audit log
Eyal's Edits	Manual post-macro	Rule-based automation	Eliminated (scoped outputs)
New Currency/Region	Full manual effort	Parameterized	Config-driven
Cycle Time	~10 business days	3-5 business days	Same-day capable
Error Detection	Post-review only	Pre-release gate	Real-time AI validation

10. Delivery Timeline

The table below maps each focus area to its delivery window, expected business milestone, and the QA gate that confirms completion before the next stage begins.

Timeline	Focus Area	Business Milestone	QA Gate
PHASE A - Live QA on PB Draft • Real-time validation as the manager edits, before PM changes begin			
Days 1-5	Live Field Validation	Real-time QA layer on PB Draft file: validates fields as the manager edits (price, CPN format, descriptions). Alerts immediately on errors, inconsistencies, or anomalies before PMs begin their changes.	<i>Live QA active</i>
PHASE B - Automation • Same process, zero manual effort			
Days 6-7	Setup	Environments ready, data access confirmed, repo initialized	<i>Data access validated</i>
Days 8-10	QA Automation Script	All 15 pre-release validation rules scripted and passing automatically	<i>QA script passing</i>
Days 11-14	ACM Duplicate Engine	Python detects and resolves ACM CPN conflicts without manual intervention	<i>Zero duplicate CPNs</i>
Days 15-19	NA Uplift Automation	All regional pricing rules applied programmatically across all categories	<i>NA uplift validated</i>
Days 20-24	Agile Picklist Sync	Discount Level and CPN Type validated automatically against live Agile	<i>Picklist alignment confirmed</i>
Days 25-28	Pricing Dashboard	Visual verification tool with filters, price charts, and uplift overlay live	<i>Dashboard validated</i>
Days 28-30	Phase B UAT	Full pipeline run end-to-end; all outputs match manual baseline exactly	<i>≥95% accuracy vs. manual</i>
PHASE C - Transformation • Beyond Excel: integrated, scalable pipeline			
Days 30-32	Unified Data Model	Single source of truth replaces dual PM PB / Macro PB file structure	<i>Data model validated</i>
Days 33-35	API Integration	Agile and Oracle connected via live API; manual entry fully eliminated	<i>API integration tested</i>
Days 36-37	Pipeline Engine	Scheduled ETL pipeline replaces VBA macro; on-demand generation live	<i>Pipeline stable E2E</i>
Days 38-40	AI Anomaly Detection	Pattern-based pricing anomaly engine trained and flagging outliers	<i>Anomaly engine validated</i>
Days 40-42	UAT & Go-Live	Business sign-off received; full pipeline deployed across all regions	<i>Go-live approved</i>

11. Success Criteria

The initiative will be considered successful when the following outcomes are verifiably achieved:

Phase A - Live QA Active	Real-time validation running on the PB Draft file. Every field edit validated instantly; pattern-based anomaly alerts firing before PMs begin their changes.
Phase B - QA Automated	All 15 validation checks pass automatically before every macro run. Zero manual QA intervention required.
Phase B - ACM Accuracy	ACM duplicate CPNs detected and resolved programmatically. No manual conflict resolution in any quarter.
Phase B - Script Stability	Refactored Python ACM script runs on pandas 2.0+ with externalized product mappings. No code changes for new products.
Phase B - Pricing Dashboard	Visual pricing verification tool live: filter by model, view price charts, overlay NA uplift, compare versions. PMs can verify strategy in minutes.
Phase C - Pipeline Live	Full quarterly Price Book suite generated from a single pipeline run without spreadsheet-based manual steps.
Phase C - System Integration	Agile picklist sync and Oracle ACM updates are completed automatically as part of the standard release pipeline.
Phase C - On-Demand	Price Book generation can be triggered at any time, not constrained to the quarterly cycle window.
Phase C - AI Anomaly Engine	Pattern-based anomaly detection flags pricing outliers (suffix patterns, category drift, ACM ratio deviations) with plain-English explanations before release.
Both Phases - Zero Regression	All generated outputs match or exceed current accuracy. No pricing discrepancies introduced by automation.